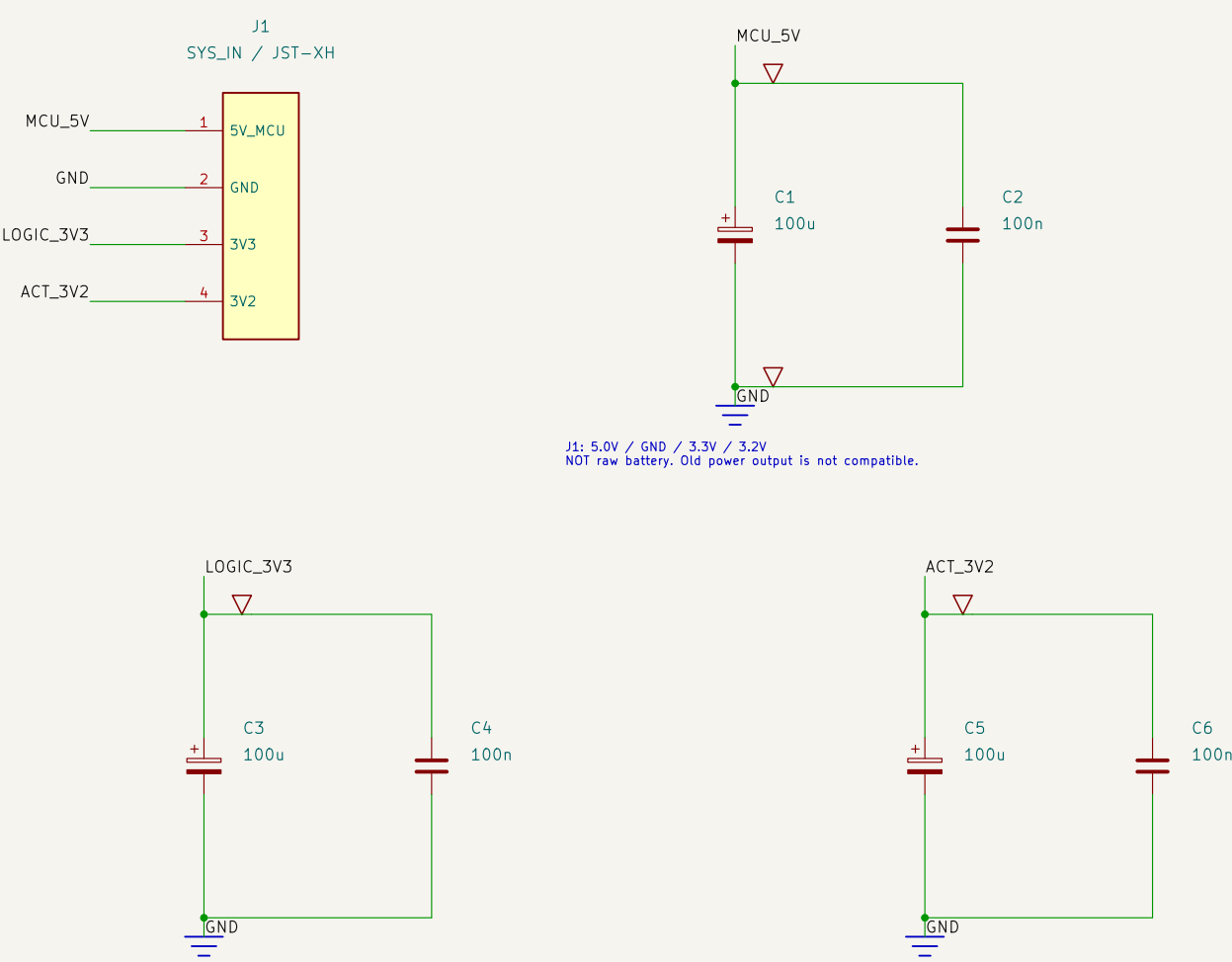


KEYCHAIN KREATURES / MAIN BOARD

ONE-SHEET CIRCUIT VIEW | C.5: Landscape screen / rear MCU / compact through-hole placement | Engineering prototype

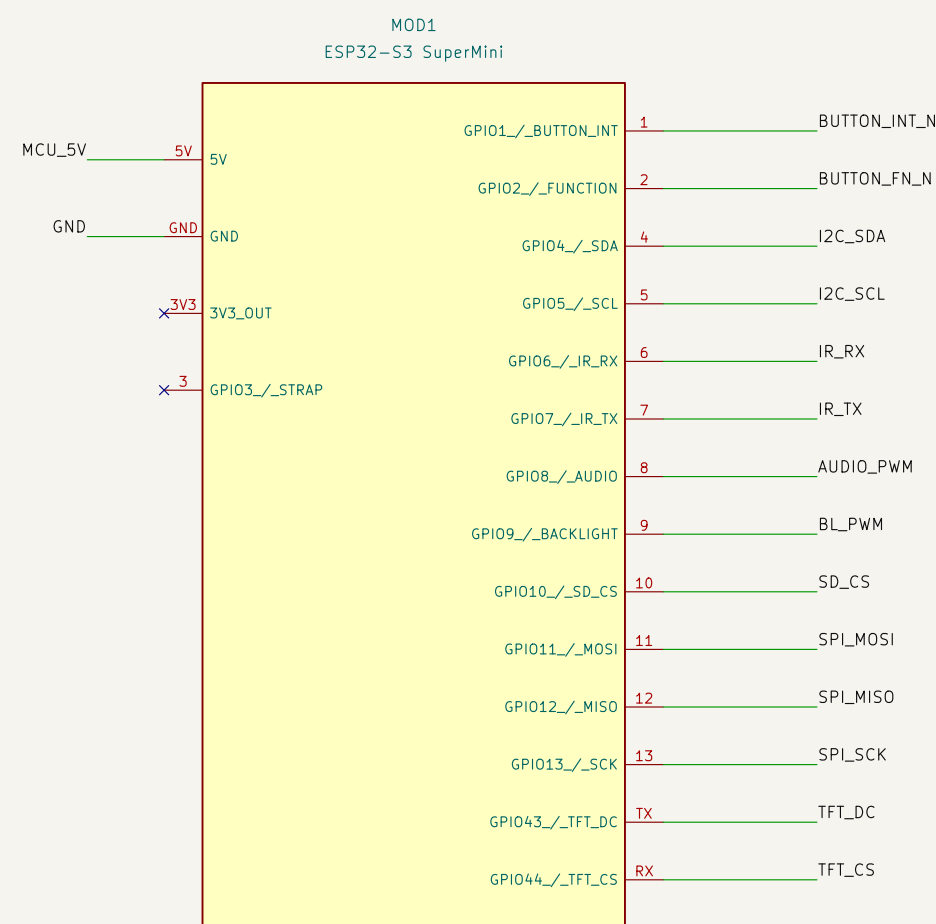
POWER INPUT + LOCAL BYPASS

Regulation / charging stay on the separate power board.



ESP32-S3 SUPERMINI / SOCKET

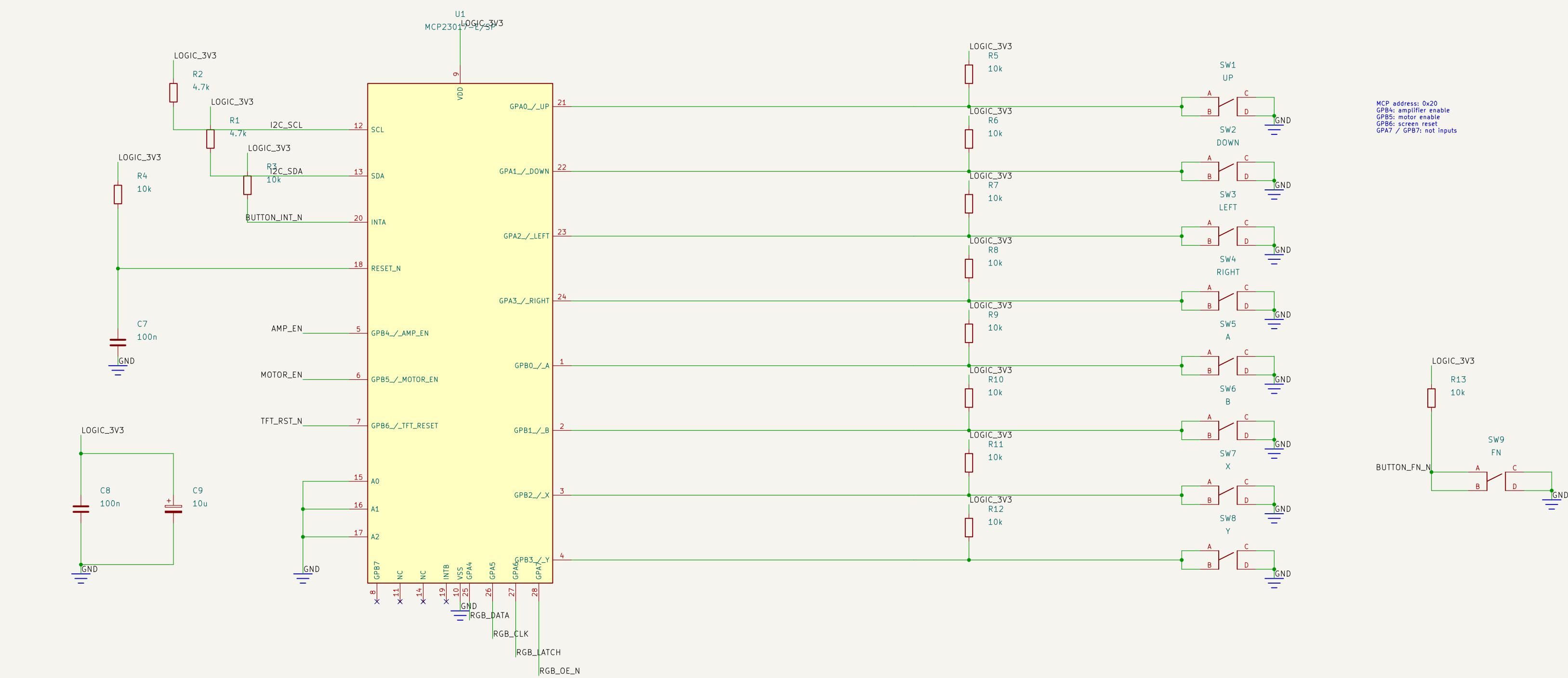
Header signals shown by GPIO; pad names match supplied footprint



USB service: remove socketed MCU first.
3V3 OUT is intentionally not tied to the external 3V3 rail

BUTTONS + GPIO EXPANDER

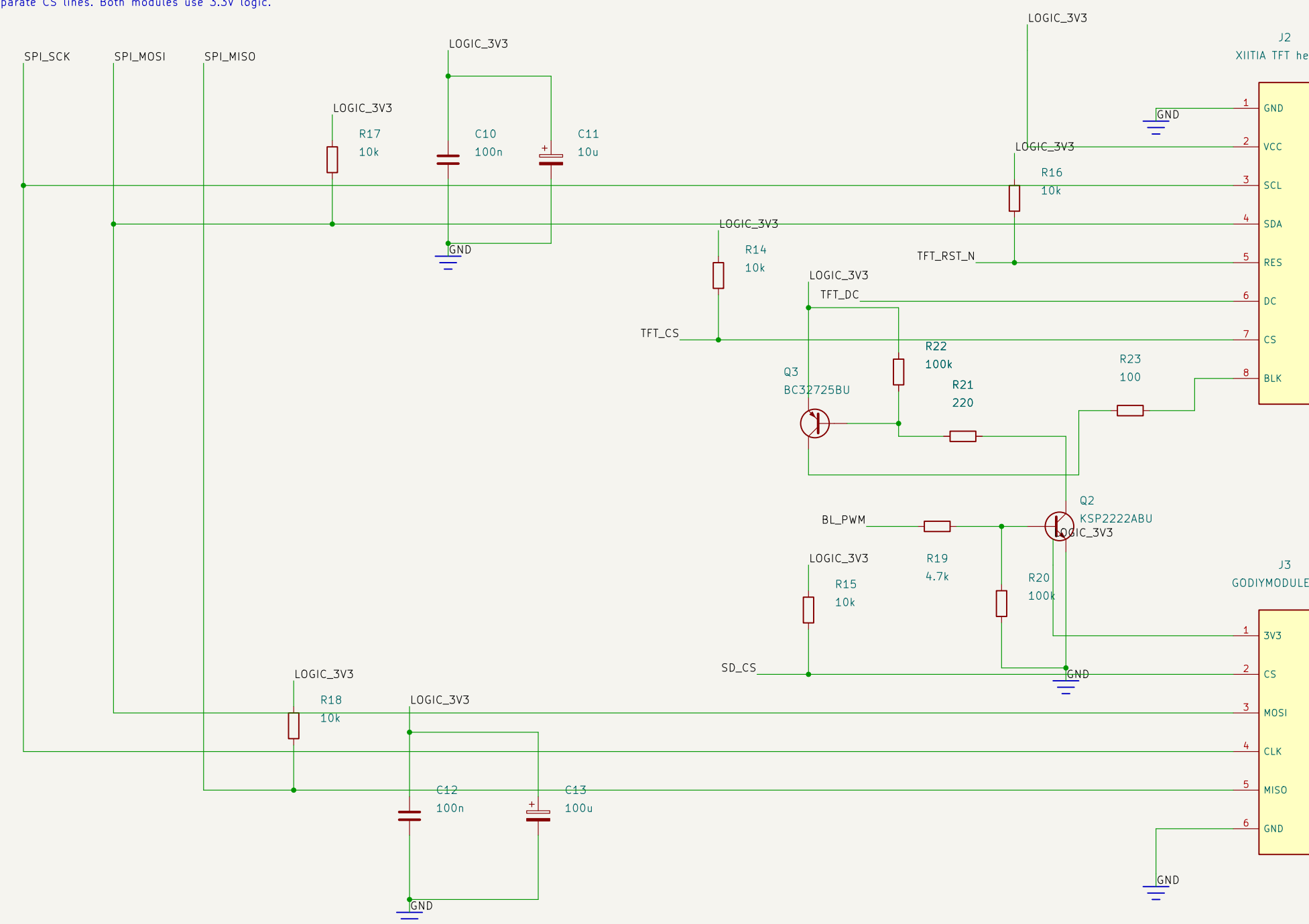
Each switch is wired to its pull-up and input. A/B are one contact; C/D are the other



MCP address: 0x20
GPB4: amplifier enable
GPB5: motor enable
GPB6: screen reset
GPA7 / GPB7: not input

SCREEN + microSD + BACKLIGHT

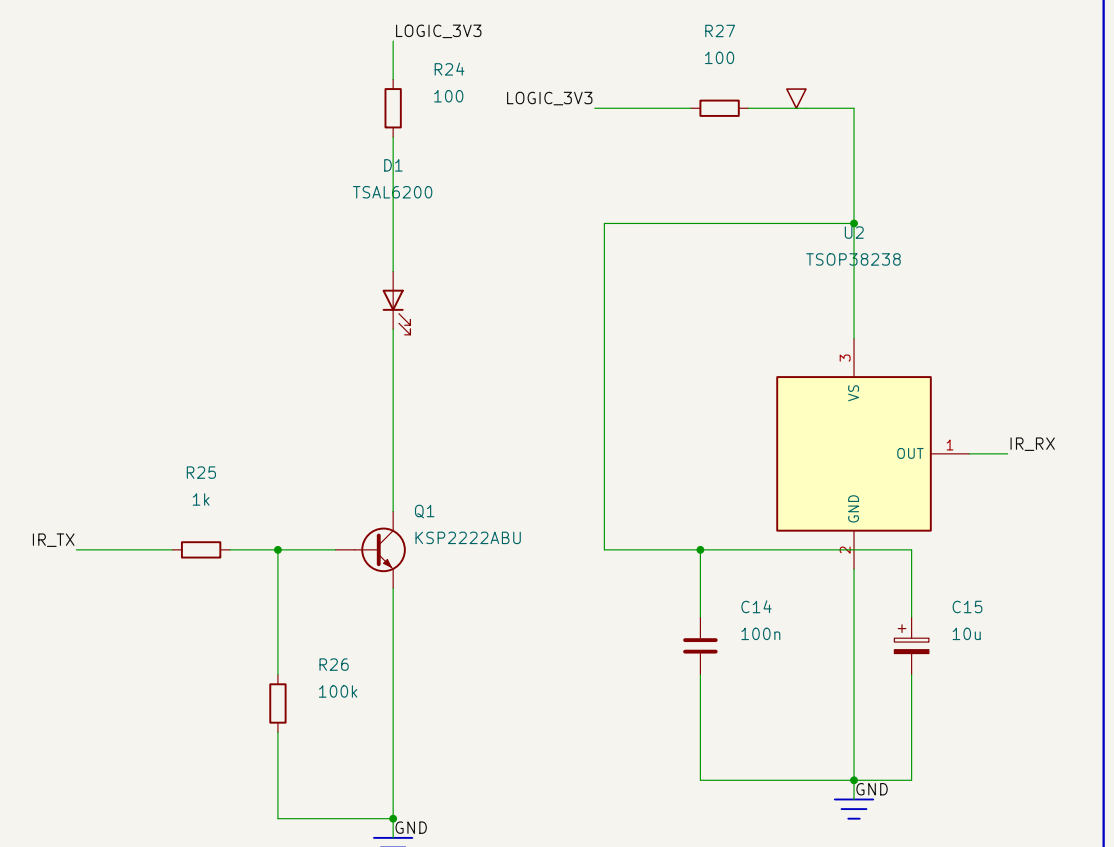
Shared SPI clock / MOSI. Separate CS lines. Both modules use 3.3V logic



R23 = conservative bring-up current limit.
Confirm BLK circuitry before raising brightness

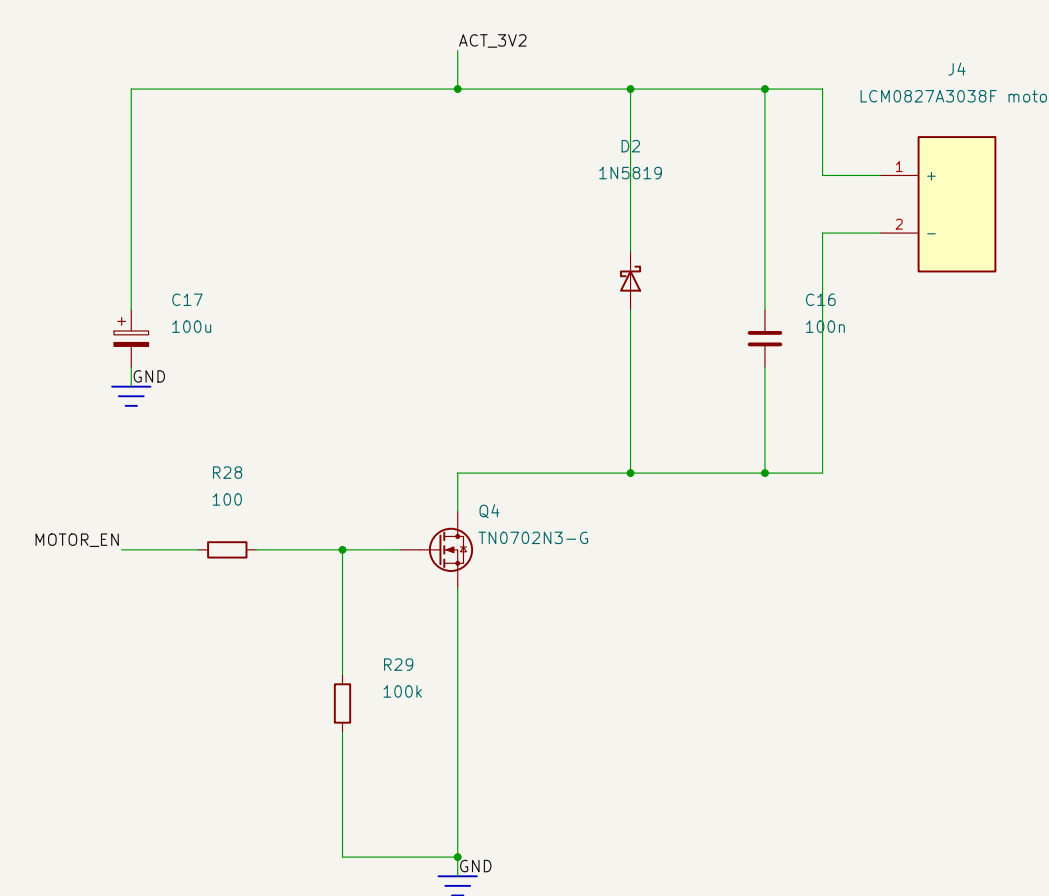
IR TRANSMIT + RECEIVE

38kHz remote-control IR, not sub-GHz RF



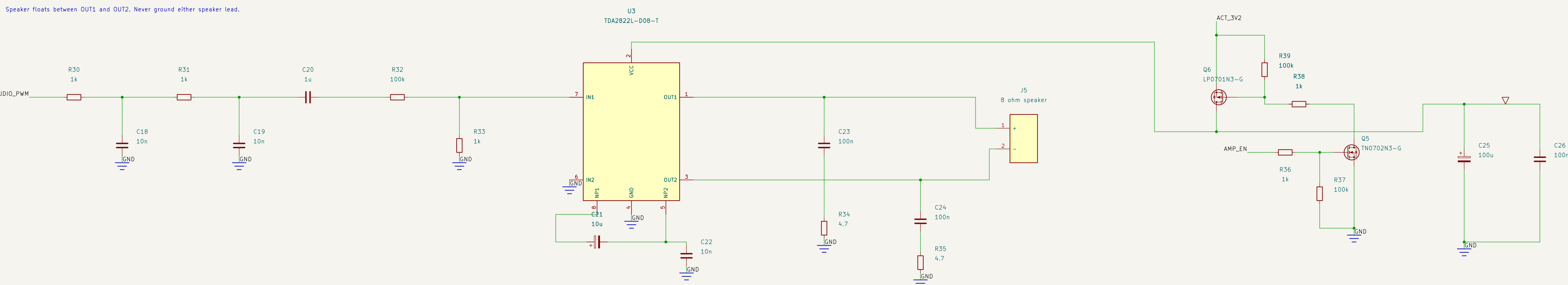
VIBRATION MOTOR

Flyback diode + low-side MOSFET.



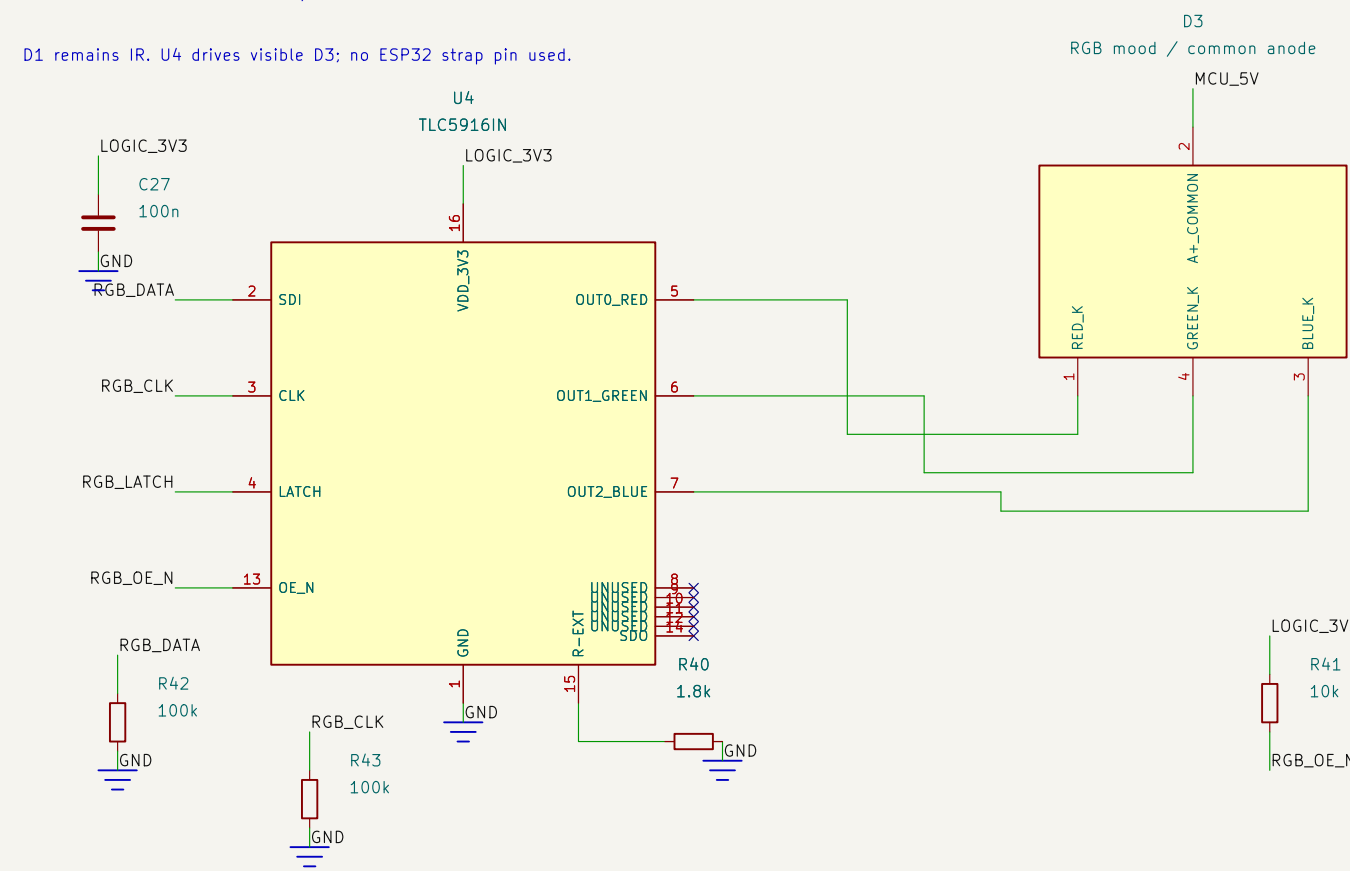
AUDIO / PWM FILTER -> BRIDGE AMPLIFIER -> SPEAKER

Speaker floats between OUT1 and OUT2. Never ground either speaker lead.



RGB MOOD LED / CONSTANT CURRENT

D1 remains IR. U4 drives visible D3; no ESP32 strap pin used.



GPA4=DATA; GPA5=CLK; GPA6=LATCH; GPA7=OE_N
Default blanked; latch 0 before enabling. Approx. 10.4mA/channel